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In [2]: # Name - Rohan Pise
        # Rollno. - 01
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In [5]: import tensorflow as tf
        from sklearn.model_selection import train_test_split
        from sklearn.preprocessing import StandardScaler
        from sklearn.datasets import load_breast_cancer

        # Load the breast cancer dataset
        data = load_breast_cancer()
        X_train, X_test, y_train, y_test = train_test_split(data.data, data.target, test_si

        # Scale the input features
        scaler = StandardScaler()
        X_train = scaler.fit_transform(X_train)
        X_test = scaler.transform(X_test)

        # Build the model
        model = tf.keras.models.Sequential([
            tf.keras.layers.Dense(1, activation='sigmoid', input_shape=(X_train.shape[1],))
        ])

        # Compile the model
        model.compile(optimizer='adam', loss='binary_crossentropy', metrics=['accuracy'])

        # Train the model
        model.fit(X_train, y_train, epochs=5)

        # Evaluate the model
        test_loss, test_accuracy = model.evaluate(X_test, y_test)
        print("Accuracy:", test_accuracy)
```

Epoch 1/5

C:\Users\vidhi\anaconda3\Lib\site-packages\keras\src\layers\core\dense.py:87: UserWarning: Do not pass an `input\_shape`/`input\_dim` argument to a layer. When using Sequential models, prefer using an `Input(shape)` object as the first layer in the model instead.

```
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
15/15 ————— 1s 3ms/step - accuracy: 0.6290 - loss: 0.6340
Epoch 2/5
15/15 ————— 0s 4ms/step - accuracy: 0.7904 - loss: 0.5335
Epoch 3/5
15/15 ————— 0s 3ms/step - accuracy: 0.8539 - loss: 0.4826
Epoch 4/5
15/15 ————— 0s 4ms/step - accuracy: 0.8508 - loss: 0.4420
Epoch 5/5
15/15 ————— 0s 3ms/step - accuracy: 0.8996 - loss: 0.3966
4/4 ————— 0s 11ms/step - accuracy: 0.8856 - loss: 0.3728
Accuracy: 0.8859649300575256
```

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In [ ]:
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